RAJALAKSHMI ENGINEERING COLLEGE

Harini P 251701015-2025-CSD

H2

Started on	Monday, 28 September 2026, 4:01 PM
State	Finished
Completed on	Monday, 28 September 2026, 9:18 PM
Time taken	5 hours 17 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int majority(int a[],int l,int r){
    if(l==r)
    return a[l];
    int mid=(l+r)/2;
    int x=majority(a,l,mid);
    int y=majority(a,mid+1,r);
    if(x==y)
    return x;
    int cx=0,cy=0;
    for(int i=l;i<=r;i++){
        if(a[i]==x)
        cx++;
        if(a[i]==y)
        cy++;
    }
    if(cx>cy)
    return x;
    else
    return y;
}
int main(){
    int a;
    scanf("%d",&a);
    int arr[a];
    for(int i=0;i<a;i++)
    scanf("%d",&arr[i]);
    printf("%d",majority(arr,0,a-1));
}
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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